```
clc;
clear;
close all;

%% IIR Filter Coefficients
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

N = 50;
x = [1 zeros(1,N-1)];  % Unit impulse
n = 0:N-1;

%% 1. Direct Form-I and Direct Form-II
y1 = filter(b,a,x);
y2 = filter(b,a,x);

%% 2. Cascade Realization
[sos,g] = tf2sos(b,a);

yc = g * x;

for k = 1:size(sos,1)
    yc = filter(sos(k,1:3),sos(k,4:6),yc);
end

%% 3. Parallel Realization
[r,p,k] = residuez(b,a);

% Initialize parallel output with direct term
yp = filter(k,1,x);

for i = 1:length(r)
    yp = yp + filter(r(i),[1 -p(i)],x);
end

% Remove negligible imaginary numerical components
if isreal(b) && isreal(a)
    y1 = real(y1);
    y2 = real(y2);
    yc = real(yc);
    yp = real(yp);
end

%% 4. Display Coefficients
disp('Numerator coefficients b =');
```

```
Numerator coefficients b =
```

```
disp(b);
```

```
    0.0048    0.0193    0.0289    0.0193    0.0048
```

```
disp('Denominator coefficients a =');
```

```
Denominator coefficients a =
```

```
disp(a);
```

```
    1.0000   -2.3695    2.3140   -1.0547    0.1874
```

```
disp('Second Order Sections (Cascade) =');
```

```
Second Order Sections (Cascade) =
```

```
disp(sos);
```

```
    1.0000    2.1551    1.0000    1.0000   -1.0486    0.2962
    1.0000    1.8657    1.0000    1.0000   -1.3209    0.6327
```

```
disp('Cascade Gain =');
```

```
Cascade Gain =
```

```
disp(g);
```

```
    0.0048
```

```
disp('Residues (Parallel) =');
```

```
Residues (Parallel) =
```

```
disp(r);
```

```
  -0.2698 + 0.1343i
  -0.2698 - 0.1343i
   0.2594 - 0.8024i
   0.2594 + 0.8024i
```

```
disp('Poles (Parallel) =');
```

```
Poles (Parallel) =
```

```
disp(p);
```

```
   0.6604 + 0.4433i
   0.6604 - 0.4433i
   0.5243 + 0.1459i
   0.5243 - 0.1459i
```

```
disp('Direct term =');
```

```
Direct term =
```

```
disp(k);
```

```
    0.0256
```

```
%% 5. Error and Percentage Deviation
E = [
    0
    max(abs(y1-y2))
    max(abs(y1-yc))
    max(abs(y1-yp))
];

reference = max(abs(y1));

if reference > 0
    D = E/reference*100;
else
    D = zeros(size(E));
end

Structure = {'DF-I'; 'DF-II'; 'Cascade'; 'Parallel'};

T = table(Structure,E,D, ...
    'VariableNames', ...
    {'Structure','Maximum_Error','Deviation_percent'});

disp('Comparison Table:');
```

```
Comparison Table:
```

```
disp(T);
```

Structure	Maximum_Error	Deviation_percent
{'DF-I'}	0	0
{'DF-II'}	0	0
{'Cascade'}	1.9221e-15	8.242e-13
{'Parallel'}	1.9811e-15	8.4949e-13

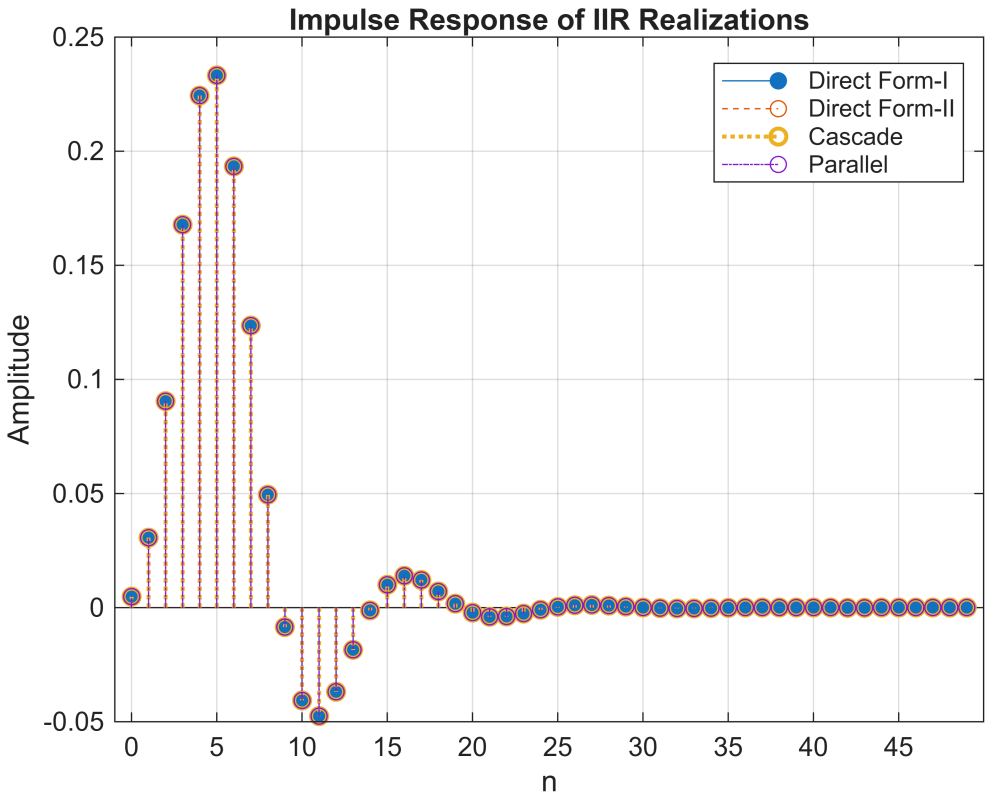
```
%% 6. Impulse Response Comparison
figure('Name','Impulse Response Comparison');

stem(n,y1,'filled');
hold on;
stem(n,y2,'--');
stem(n,yc,':','LineWidth',1.5);
stem(n,yp,'-.');

grid on;
xlabel('n');
```

```
ylabel('Amplitude');
title('Impulse Response of IIR Realizations');

legend('Direct Form-I','Direct Form-II', ...
       'Cascade','Parallel','Location','best');
```

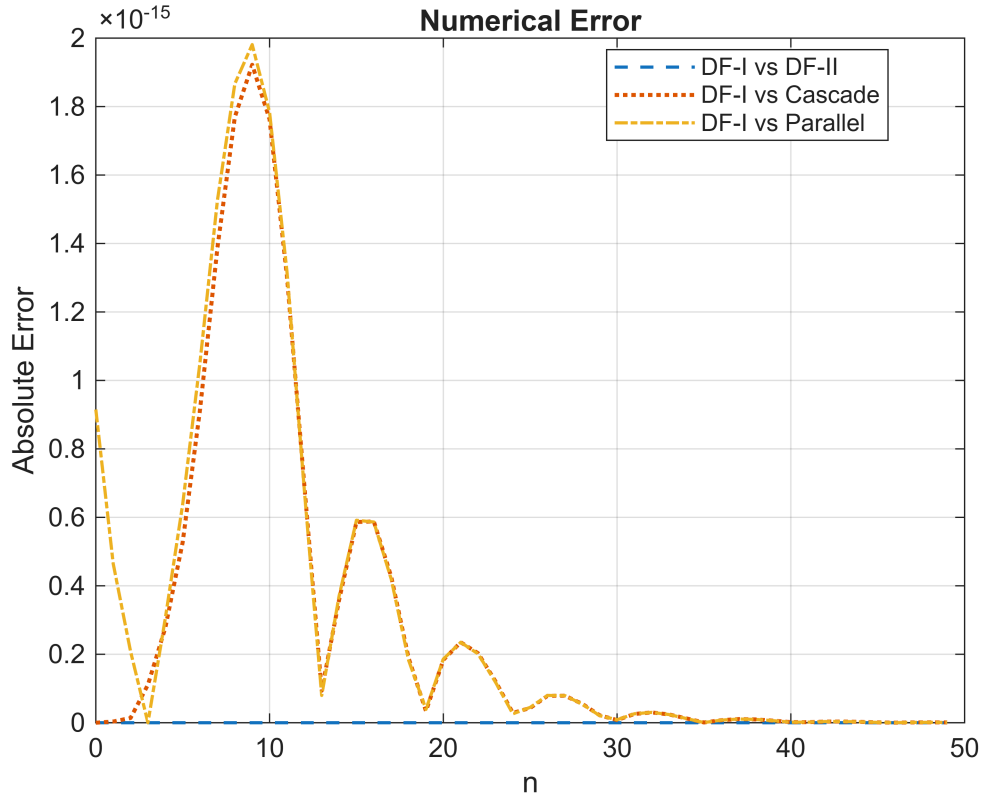



```
%% 7. Error Plot
figure('Name','Numerical Error');

plot(n,abs(y1-y2),'--','LineWidth',1.2);
hold on;
plot(n,abs(y1-yc),':','LineWidth',1.5);
plot(n,abs(y1-yp),'-.','LineWidth',1.2);

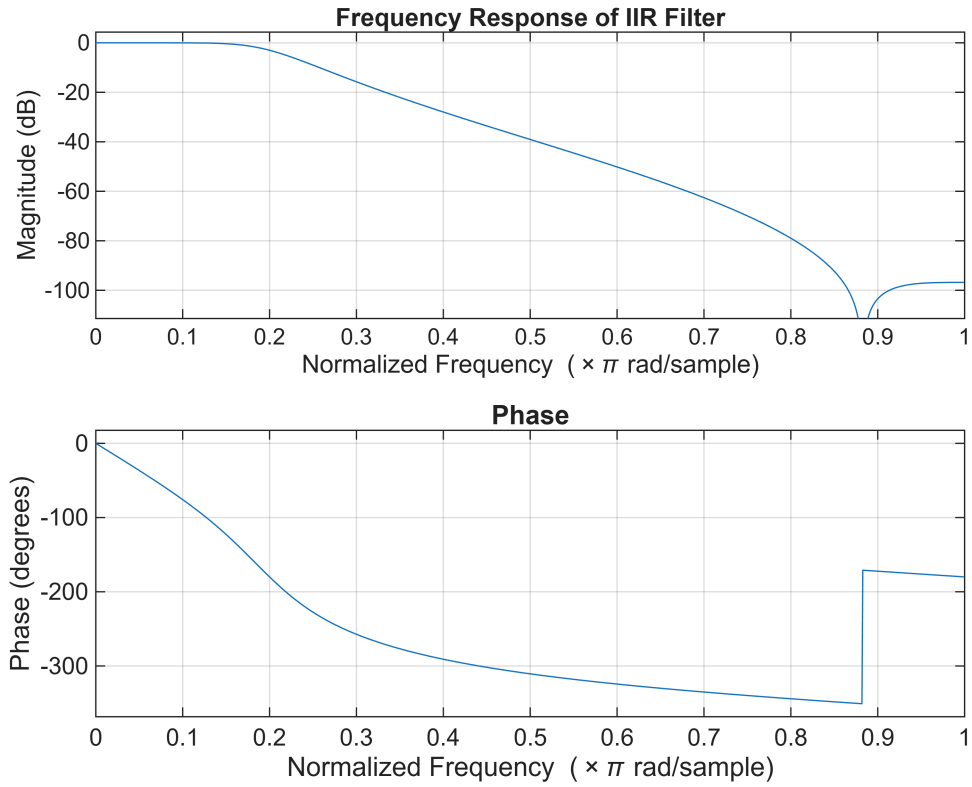
grid on;
xlabel('n');
ylabel('Absolute Error');
title('Numerical Error');

legend('DF-I vs DF-II','DF-I vs Cascade', ...
       'DF-I vs Parallel','Location','best');
```



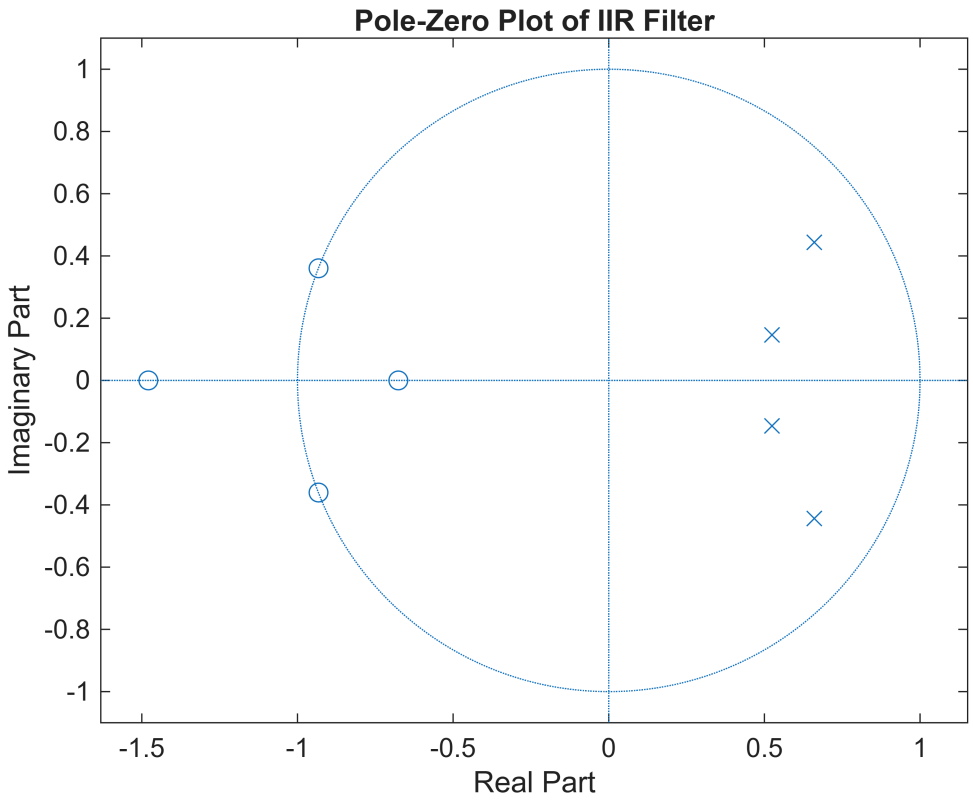

```
%% 8. Frequency Response
figure('Name','Frequency Response');

freqz(b,a,1024);
title('Frequency Response of IIR Filter');
```




```
%% 9. Pole-Zero Plot
figure('Name','Pole-Zero Plot');

zplane(b,a);
title('Pole-Zero Plot of IIR Filter');
```


Pole-Zero Plot of IIR Filter
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.5
-1
-0.5
0
0.5
1